

Nonreciprocal cancer–immune dynamics with Allee effects and immunological control

Erick Serrato García¹

¹*Departamento de Física, Universidad Autónoma Metropolitana Iztapalapa*

[Abstract en una página: describir efecto Allee (débil/fuerte), no reciprocidad en interacciones cáncer–inmune y control inmunológico adaptativo $u(x, t)$ (texto plano). Resumir hallazgos de puntos fijos inestables y patrones espaciales bajo control.]

I. INTRODUCTION

A. Cancer as a nonequilibrium system

Tumor growth is a driven, open system with matter and energy fluxes. Nonequilibrium physics provides tools to interpret multistability, thresholds, and pattern formation in tumor–immune competition.

B. Limitations of reciprocal population models

Symmetric Lotka–Volterra or logistic models miss thresholds, hysteresis, and asymmetric suppression/activation between cancer, healthy tissue, and immune cells. These limitations motivate adding Allee effects and nonreciprocal couplings.

C. Allee effects and critical thresholds

An Allee term introduces a viability threshold for cancer: below it, extinction; above it, proliferation. Weak Allee yields a soft slowdown; strong Allee creates a sharp barrier.

D. Nonreciprocal interactions in cancer–immune dynamics

Suppression of immunity by tumor cells and cytotoxic action of immune cells are not symmetric. Including nonreciprocity alters nullclines, fixed points, and transient routes.

E. Summary of contributions

- Formulate ODE/PDE models with weak/strong Allee effects, nonreciprocity, and an immunological control field $u(x, t)$.
- Characterize fixed points and linear stability; identify Allee-driven bifurcations.
- Extend to reaction–diffusion dynamics; report pattern formation and length scales.
- Evaluate control strategies (constant vs adaptive) and their cost–efficiency trade-offs.

II. MODEL

A. Population variables

- $c(t)$ or $c(\mathbf{x}, t)$: cancer cell density.
- $s(t)$ or $s(\mathbf{x}, t)$: healthy tissue density.
- $i(t)$ or $i(\mathbf{x}, t)$: immune effector density.

B. Baseline dynamics (ODE)

We start from logistic-like growth and quadratic interaction terms capturing cytotoxicity, tissue damage, and immune cooperation [2]:

$$\dot{c} = r_c c f_{\text{Allee}}(c) - \alpha c s - \beta c i - \frac{\mu}{2}(\eta i^2 + \gamma s^2), \quad (1)$$

$$\dot{s} = r_s s \left(1 - \frac{s}{k_s}\right) - \gamma c s + \delta s i^2 - \frac{\mu}{2}\alpha c^2, \quad (2)$$

$$\dot{i} = r_i i \left(1 - \frac{i}{k_i}\right) - \eta c i + \delta s^2 i - \frac{\mu}{2}\beta c^2. \quad (3)$$

C. Allee effect formulation

- **Weak Allee:** $f_{\text{Allee}}(c) = \left(\frac{c}{a} - 1\right) \left(1 - \frac{c}{k_c}\right)$ (soft threshold) [1].
- **Strong Allee:** $f_{\text{Allee}}(c) = \left(\frac{c}{a} - 1\right) \left(\frac{c-a}{k_c-a}\right)$ (hard threshold) [4].

D. Nonreciprocal interactions

Suppression of immunity by cancer ($-\eta c i$) differs from immune cytotoxicity ($-\beta c i^2$); damage to healthy tissue by cancer ($-\gamma c s$) differs from competition against cancer ($-\alpha c s^2$). These asymmetries tilt nullclines and shift bifurcations.

E. Immunological control field $u(\mathbf{x}, t)$

- Binary/parametric control: $\mu \in \{0, 1\}$ toggles variational terms.
- Adaptive field: $u(\mathbf{x}, t) = u_0 \frac{c}{i + \varepsilon}$ with cap u_{max} to prioritize high- c , low- i regions.
- Coupling: control can enter as enhanced cytotoxicity ($\beta \rightarrow \beta + u$) or as an external field in the i -equation.

F. Spatial extension (PDE)

Adding diffusion yields

$$\partial_t c = D_c \nabla^2 c + r_c c f_{\text{Allee}}(c) - c(\alpha s^2 + \beta i^2) - \mu c(\gamma s^2 + \eta i^2), \quad (4)$$

$$\partial_t s = D_s \nabla^2 s + r_s s(1 - s) - \gamma c^2 s + \delta i^2 s - \frac{\mu}{2}\alpha s c^2, \quad (5)$$

$$\partial_t i = D_i \nabla^2 i + r_i i(1 - i) + \delta s^2 i - \eta c^2 i - \frac{\mu}{2}\beta c^2 i. \quad (6)$$

Diffusion acts as a mode selector; with $D = \text{diag}(D_c, D_s, D_i)$ the linear spectrum is $\lambda(k) = \text{eig}(J - Dk^2)$.

Symbol	Description	Example value
r_c, r_s, r_i	intrinsic growth rates (cancer, healthy, immune)	5.84, 13.12, 10.92
k_c, k_s, k_i	carrying capacities	1, 1, 1
a	Allee threshold (weak/strong)	7.22 (weak), 0.1 (strong scans)
$\alpha, \beta, \gamma, \delta, \eta$	nonreciprocal interaction strengths	10.22, 7.60, 0.74, 5.40, 5.08
D_c, D_s, D_i	diffusion coefficients (spatial model)	1, 1, 1
μ	control/variational switch	$\{0, 1\}$
$u(\mathbf{x}, t)$	adaptive immunological control field	capped by u_{max}

TABLE I. Variables y parámetros del modelo. Ajustar valores según régimen weak/strong y escenarios de control.

III. MEAN-FIELD ANALYSIS

A. Fixed points

Solve $\dot{c} = \dot{s} = \dot{i} = 0$ for weak/strong Allee. Typical saddles: weak Allee $(c^*, s^*, i^*) \approx (1.1, 0.5, 0.5)$ with $\text{Re } \lambda_{\max} \sim 60$; strong Allee near $(1.0, 0.0, 0.1)$ with $\text{Re } \lambda_{\max} \sim 9$.

B. Linear stability

Jacobian $J(c^*, s^*, i^*)$; for spatial perturbations use $\lambda_j(k) = \text{eig}(J - Dk^2)$.

C. Allee-induced bifurcations

Vary a and r_c to track saddle-node or transcritical bifurcations separating extinction vs proliferation. Nonreciprocity tilts diagrams.

D. Role of nonreciprocity

Asymmetries in $\alpha, \beta, \gamma, \delta, \eta$ shift nullclines and raise/lower $\text{Re } \lambda_{\max}$; explored grids did not stabilize $i^* > 0$ equilibria.

E. Representative figures

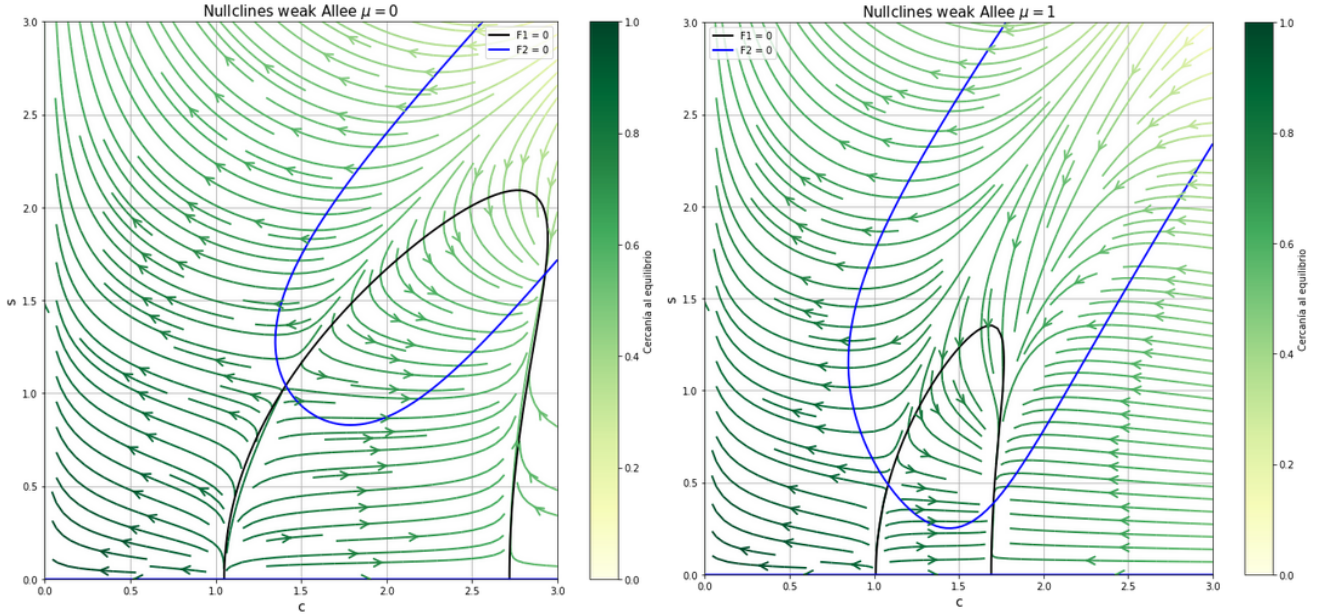


FIG. 1. Phase planes y nulclinas: efecto Allee débil con $\mu = 0$ (izq.) y $\mu = 1$ (der.).

IV. SPATIAL DYNAMICS

A. Reaction–diffusion extension

See PDE system in Sec. II; diffusion acts as selector of modos inestables.

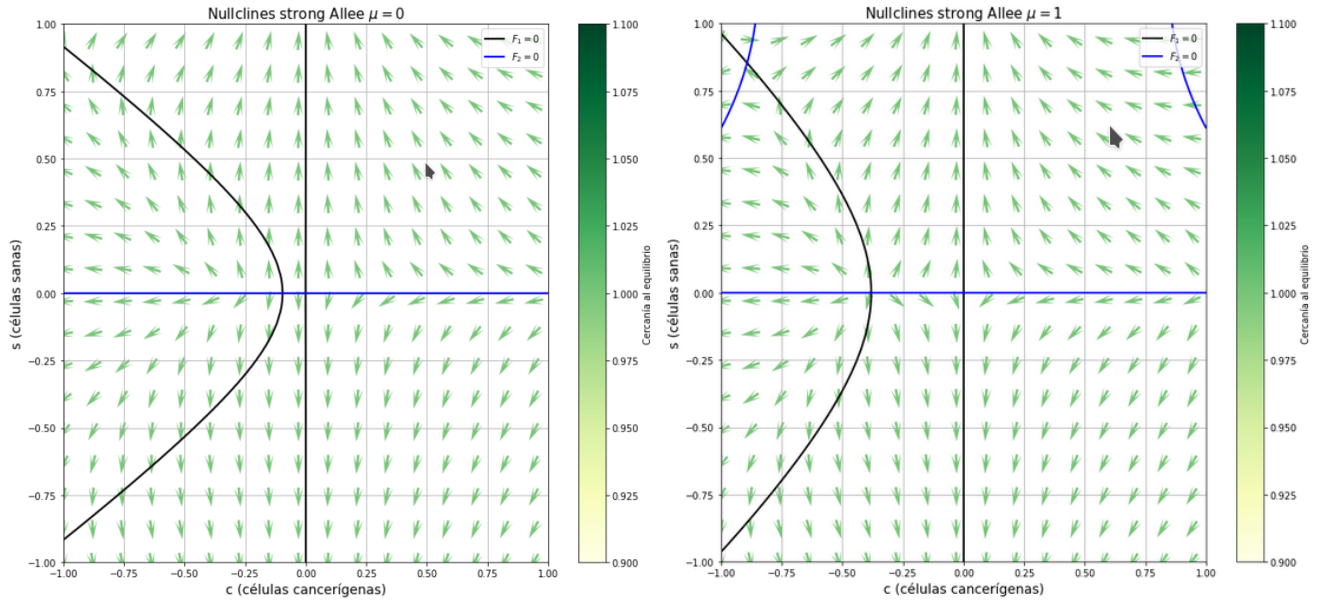


FIG. 2. Phase planes para Allee fuerte con $\mu = 0$ (izq.) y $\mu = 1$ (der.).

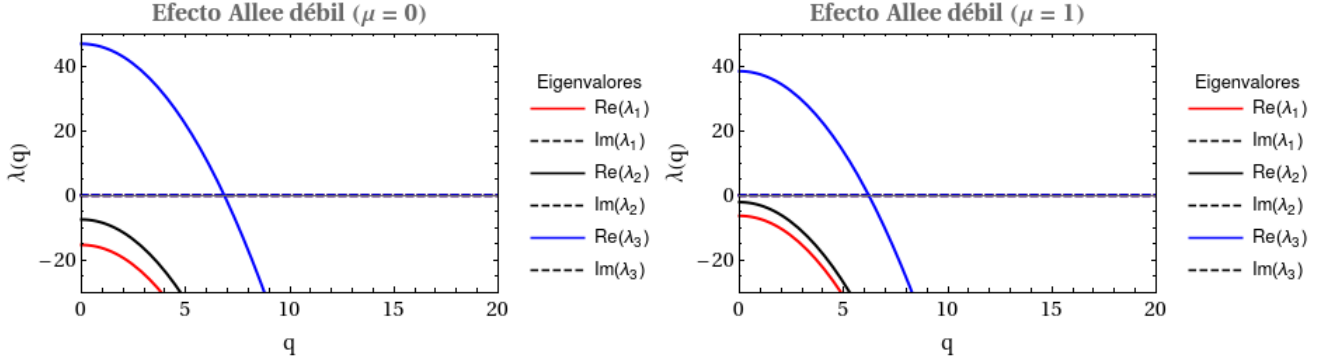


FIG. 3. Espectros lineales (Allee débil): $\mu = 1$ atenúa modos de alta frecuencia pero no estabiliza $k \approx 0$.

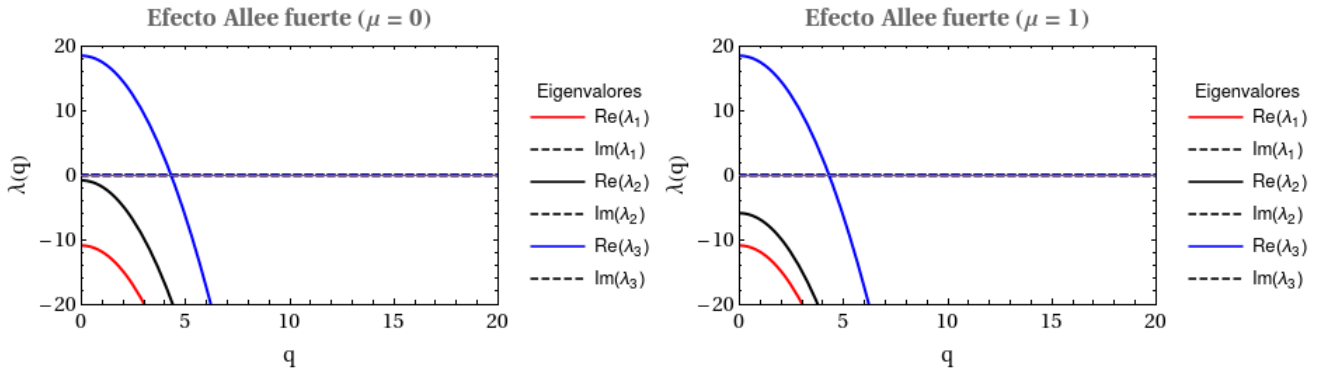


FIG. 4. Espectros lineales (Allee fuerte): $\mu = 1$ reduce la banda inestable en altas frecuencias.

B. Numerical methods

FEniCS 2D, malla uniforme (~ 200 nodos por lado), $\Delta t \sim 5 \times 10^{-3}$, $T \sim 2$. Condiciones iniciales aleatorias en rangos factibles.

C. Pattern formation

Patrones para weak/strong y $\mu = 0/1$; se evalúan espectros de potencia y longitudes de correlación $\xi(t)$.

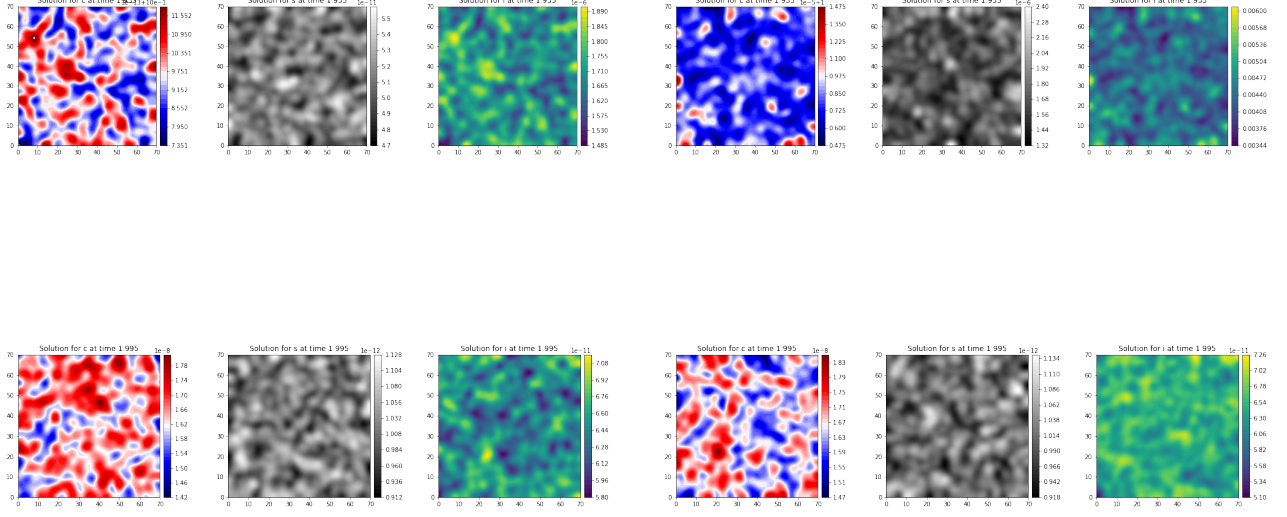


FIG. 5. Patrones espaciales de c, s, i en $T \approx 1.9$: fila superior (Allee débil) $\mu = 0$ y $\mu = 1$; fila inferior (Allee fuerte) $\mu = 0$ y $\mu = 1$.

V. IMMUNOLOGICAL CONTROL

A. Control formulation

Represent control via μ (variational switch) and an adaptive field

$$u(\mathbf{x}, t) = \min\left(u_{\max}, u_0 \frac{c}{i + \varepsilon}\right)$$

that prioritizes high- c , low- i regions. It can enhance β (cytotoxicity) or enter as a source in the immune equation.

B. Control strategies

- Constant vs adaptive dosing.
- Spatial targeting: stronger control where c is high and i is low.
- Timing protocols: pulsed vs continuous application.

C. Efficiency and cost

Cost functional (drug load $\int u d\mathbf{x} dt$) vs benefit (peak reduction, time to extinction, total tumor burden $\int c dt$, correlation length of c). Compare $\mu = 0, 1$ and adaptive u ; report Pareto curves cost vs suppression.

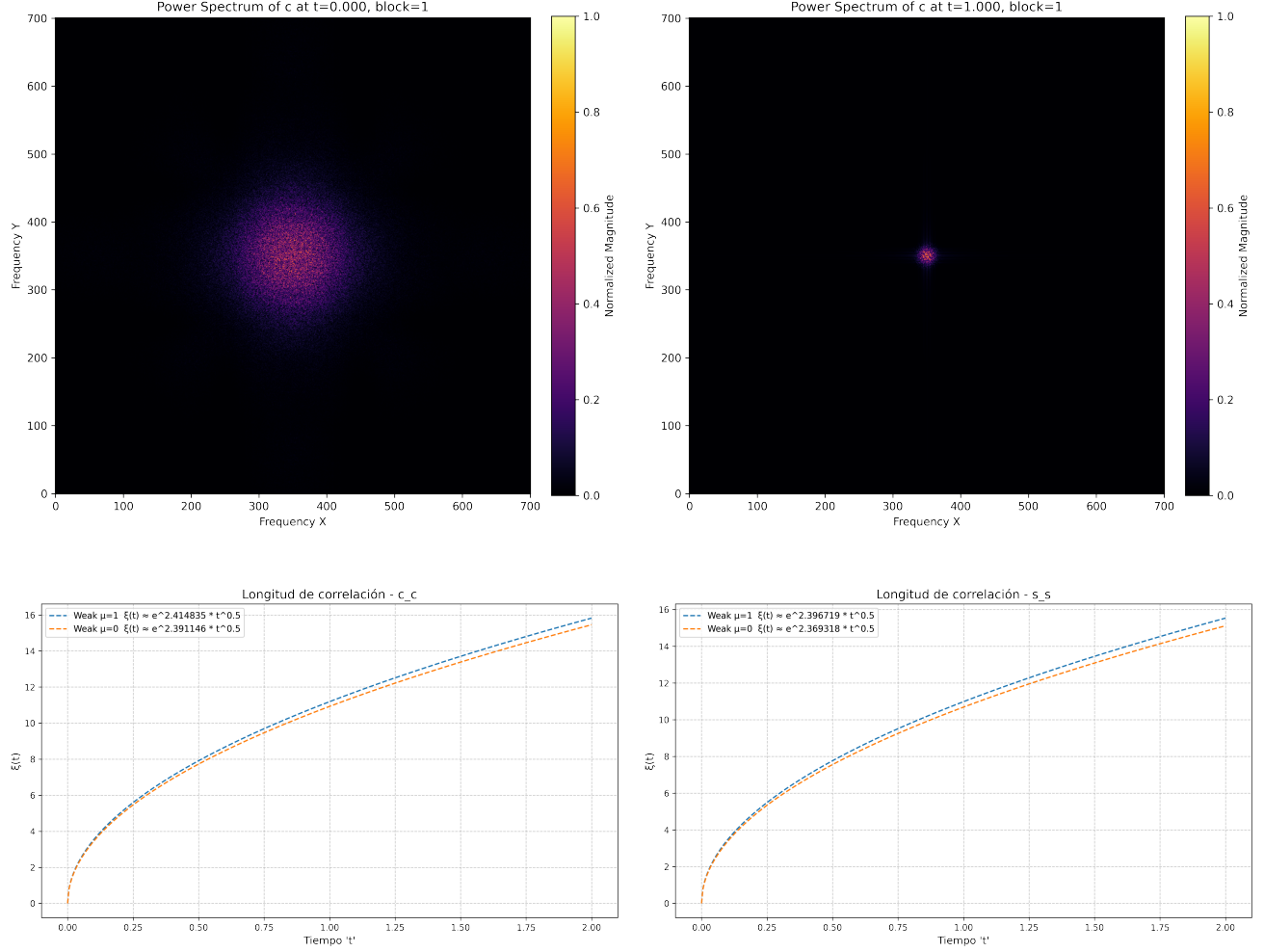


FIG. 6. Arriba: espectro de potencia del campo c al inicio y al final ($\mu = 1$, Allee débil). Abajo: evolución de la longitud de correlación $\xi(t)$ para c_c y s_s .

Protocol	Peak c reduction	Time to extinction	Cost (relative)
$\mu = 0$ (no control)	baseline	–	0
$\mu = 1$ (binary)	↓ moderate	shorter vs baseline	medium
Adaptive u (weak Allee)	↓ strong	shortest	medium–high
Adaptive u (strong Allee)	↓ strong	shortest	medium–high

TABLE II. Table II (qualitative): control efficiency vs cost. Replace with measured metrics when available.

VI. DISCUSSION

A. Physical interpretation of the results

Allee thresholds act as energy-like barriers; nonreciprocity skews flows and transient routes. Control μ or u can steer trajectories below the Allee threshold to favor extinction.

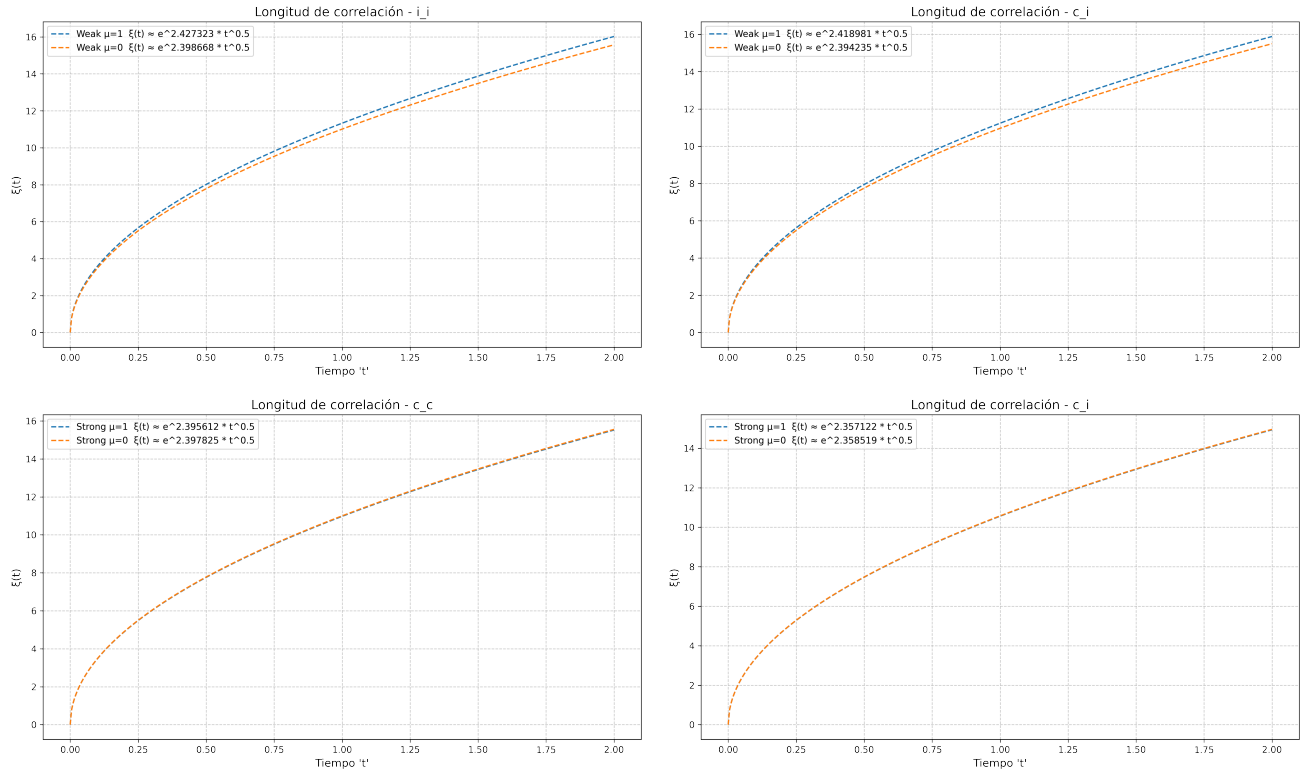


FIG. 7. Correlaciones adicionales: $\xi(t)$ para i_i y c_i (arriba), y análogos en régimen Allee fuerte (abajo).

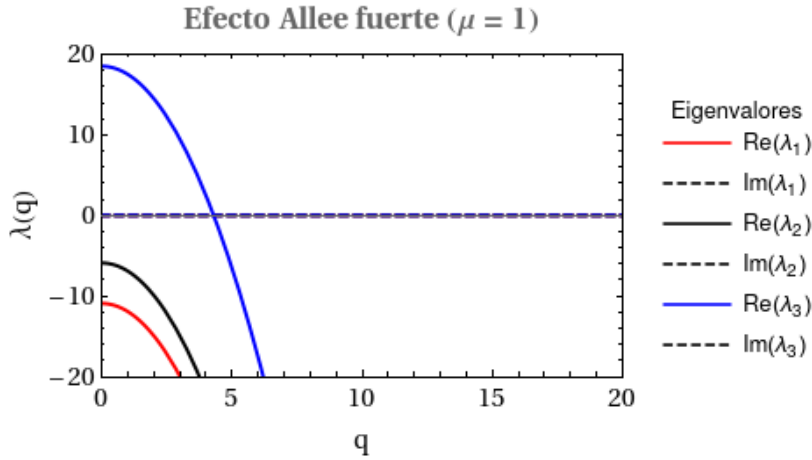


FIG. 8. Perfil espectral bajo control ($\mu = 1$) en Allee fuerte: se atenúan modos de alta frecuencia, reduciendo fragmentación.

B. Comparison with previous models

Contrast with reciprocal Lotka–Volterra and immune–tumor ODEs lacking Allee terms; highlight how bifurcations and pattern scales change with nonreciprocity and control.

C. Implications and limitations

Current scans show only unstable equilibria with $i^* > 0$ in sampled ranges; stability may emerge for lower r_c or higher rd, δ . Spatial results are short-time ($T \sim 2$); longer horizons and heterogenous domains are pending.

VII. CONCLUSIONS

- Presented a nonreciprocal cancer–immune model with weak/strong Allee effects and immunological control.
- Mean-field scans reveal only unstable fixed points in current parameter ranges; thresholds separate extinction vs proliferation.
- Spatial simulations show control-dependent pattern organization; adaptive u is expected to improve containment.
- Future work: refine parameter sweeps to seek stable attractors, optimize control protocols, and extend to longer simulations and heterogeneous geometries.

-
- [1] J. D. Murray, *Mathematical Biology I* (Springer, 2002).
 - [2] M. Delitala y T. Lorenzi, *Math. Models Methods Appl. Sci.* **17**, 1775 (2007).
 - [3] R. A. Gatenby y P. K. Maini, *Cancer Res.* **64**, 1563 (2004).
 - [4] P. Gerlee y J. Näslund, *Bull. Math. Biol.* **84**, 110 (2022).
 - [5] K. Böttger, L. H. Chao, M. O. Höft, y A. Deutsch, *PLoS Comput. Biol.* **11**, e1004369 (2015).
 - [6] A. M. Turing, *Philos. Trans. R. Soc. B* **237**, 37 (1952).